

Thursday, Bartcock F17, student meeting
2pm

- Short updates 3-4 minutes
- Good time to introduce people
(i.e. They know xxx and this might be helpful)



- What is CIRC and KRON photometry?

Qns in red

- What is turnover in stellar formation? (SFMS is linear then plateaus)

H α probes

$\approx 10^1$ Myr SFH?

Infrared probes

10^7 Myr?

- What is SFMS? (Star-Forming Main Sequence)

- Main sequence scales $\propto (1+z)$ (What does this mean?)

- What is IFU? - What is an APT file? Illustris? TNG?

Diff wavelengths probe diff SFH?

Qns I need help for
non-SED

- More ground truth sources α -Fe looks too low

Experience with spectrophot in general?

for Lega-C catalog? Working with Lega-C spectroscopically calibrated?

- More help with COSMOS photometry calibration (Laigle offset?)

- Should I incorporate metallicity history evolution? More free priors for SED fitting?

- I should make KL divergence plots?

- More help for sensible quiescent priors
- More advice for dust (e.g. emission in quiescent?)

What other parameters can I fit for

α -enhanced passive quiescent galaxies that is poorly

contained?

↳ Consideration of AGN?

- Is there fitting I can do
outside of lego-C?
Like JWST?

- JWST masking?

Automated
way to
print model?

- Talk to Jonah about
quiescent!

↳ Add Jonah and post
model assumptions

- Huge amount of ex situ
mass once star formation
shuts down

Grand $\log_{10} (M_{*} / M_{\odot}) \approx 10.8?$
Extra mass all come from mergers

↳ Stellar / luminosity
mass, not total

Stacking analysis / SDSS

for better grand truth

α -enhancement not surprising

= Summarize statistics

α / Fe $\rightarrow$ stellar mass

Tso vs stellar mass

Relative trend can be constrained
even if you only have distribution

★ List parameters in
in Slack ★

Spaxel 3D spectral cube

visualiser (!?) for galaxy?

IFU is unbiased
versus slit spectroscopy

Bulge - disk separation
and decomposition?

↳ Super hard

→ Degree of freedom

↳ $T(M_{SFR}, \delta, 3)$

alw 3

Student + SFRH → physically
motivated
rising star formation
rate

★
Strong impact
on stellar
age

Define higher
prior where you
expect physically
more star formation

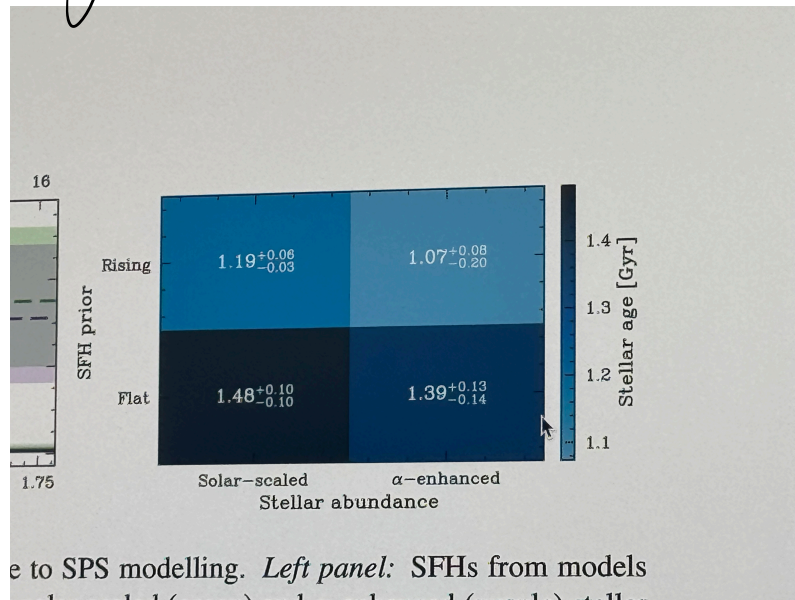
Got him to send
report

{ Mg β absorption line is
main constraint for α/Fe

Ca I can be masked b/c
of potential IGM contamination

★ Generally age estimates
get older for non α -enhanced
spectra

- Try IFU fitting
(slightly more complicated)



SFH Prior has huge dependence!
↳ Look into rising centrality prior★

prosper utik github

- He made
function to define age bins
more explicitly (but probably not
helpful)

- His rising continuity prior
comes from another paper

- Can I generate
the median, best fit, q/b q/b^4 spectrum
output

- Data reduction matters ★

- SFH for mergers could

be super messy
(you can't see this from SFH)

- GN-z11 famous galaxy?

- possible progenitor?

• α - enhanced

= Better formation timescale constraint

= More confidence on physical
formation mechanism

• Dust $\Rightarrow$ He uses Fick Corro
(modified Calzetti)

★ Given infrared photometry;

I should look into tuning
on dust emission

★ AGN torus emission

could contribute infrared emission

★ Jonah sent me his plots
and his 1st year PhD report

★ What is fagn?

★ His AGN torus parameter
is uniform in log space?

★ His redshift is a free
parameter with

★ photometry manually changed

mass and dust constraints
in his example fit

His tau dust 2 is around 0.77

★ Infrared data

↳ ① AGN torus

② Dust emission

} More physical
model